Module 03 • Control Systems & Safety

Irrigation Decision Engine & Actuator Safety Interlocks

IRRIGATEWISE Hydrogel-Based Smart Irrigation System Specification

Irrigation Decision Engine & Actuator Safety Interlocks

1. Dual-Mode Evaluation Architecture

IRRIGATEWISE features a dual-engine architecture implemented in `backend/services/irrigation_decision_engine.py`. This enables simultaneous side-by-side computation of deterministic agronomic heuristics and autonomous predictive ML recommendations.



2. Mode 1: Deterministic Soil Moisture Deficit (SMD)

The baseline agronomic engine computes the instantaneous deficit according to standard FAO-56 irrigation guidelines:

> **Equation:** `SMD = max(0, θ_optimal - θ_current)`
>
> **Equation:** `Irrigation Depth (mm) = (SMD / 100) × Z_r × 1000 × \frac1η_sys`
>
> **Equation:** `Volume (L) = Irrigation Depth (mm) × Field Area (m²)`

Plain-English Formula Breakdown:

1. **Soil Moisture Deficit (SMD):**

> **Equation:** `Deficit (%) = Target Moisture (e.g. 65%) - Current Sensor Moisture (e.g. 45%) = 20%`

(How much water is missing from the soil root zone). 2. **Irrigation Depth (mm):**

> **Equation:** `Water Depth to Apply = (Deficit / 100) × Root Depth (400 mm) × (1 / Drip Efficiency (0.85))`

3. **Total Water Volume (L):**

> **Equation:** `Total Liters Needed = Depth (mm) × Field Area (m²)`

(Since 1 mm of water spread over 1 m^2 equals exactly 1 Liter).

Where: - θ_optimal: Target root-zone moisture (65%). - θ_current: Instantaneous sensor reading (%). - Z_r: Crop rooting depth (0.4 m for typical row crops). - η_sys: Drip irrigation distribution uniformity (0.85).

Limitations of Rule-Based SMD:

1. **Reactive:** The pump only fires after depletion has already occurred.
2. **Blind to Upcoming Weather:** It irrigates fully even if heavy rainfall will occur within two hours.
3. **Ignores Hydrogel Reserves:** It applies full volume despite polymers slowly releasing moisture into surrounding root micro-pores.

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3. Mode 2: Autonomous Machine Learning Predictive Engine

Mode 2 evaluates the **forecasted root zone moisture trajectory** at t + 3h (θ̂_3h) using the Gradient Boosting regressor.

Decision Algorithm:

1. **Forecast Screening:**

> **Equation:** `Trigger Condition = (θ̂_3h < θ_trigger)`

Where θ_trigger = 46.0%. If predicted moisture at 3 hours will breach stress thresholds, irrigation is scheduled proactively. 2. **Rainfall Suppression Protocol:**

> **Equation:** `If P_forecast ≥ 5.0 mm ⇒ SUPPRESS IRRIGATION`

When precipitation is imminent, the system cancels the cycle, conserving 100% of irrigation water and avoiding waterlogging or nutrient leaching. 3. **Hydrogel Volume Modulation:** When hydrogel amendment is present (50 g/m^2), moisture desorption kinetics reduce the gross water required to restore root zone potential:

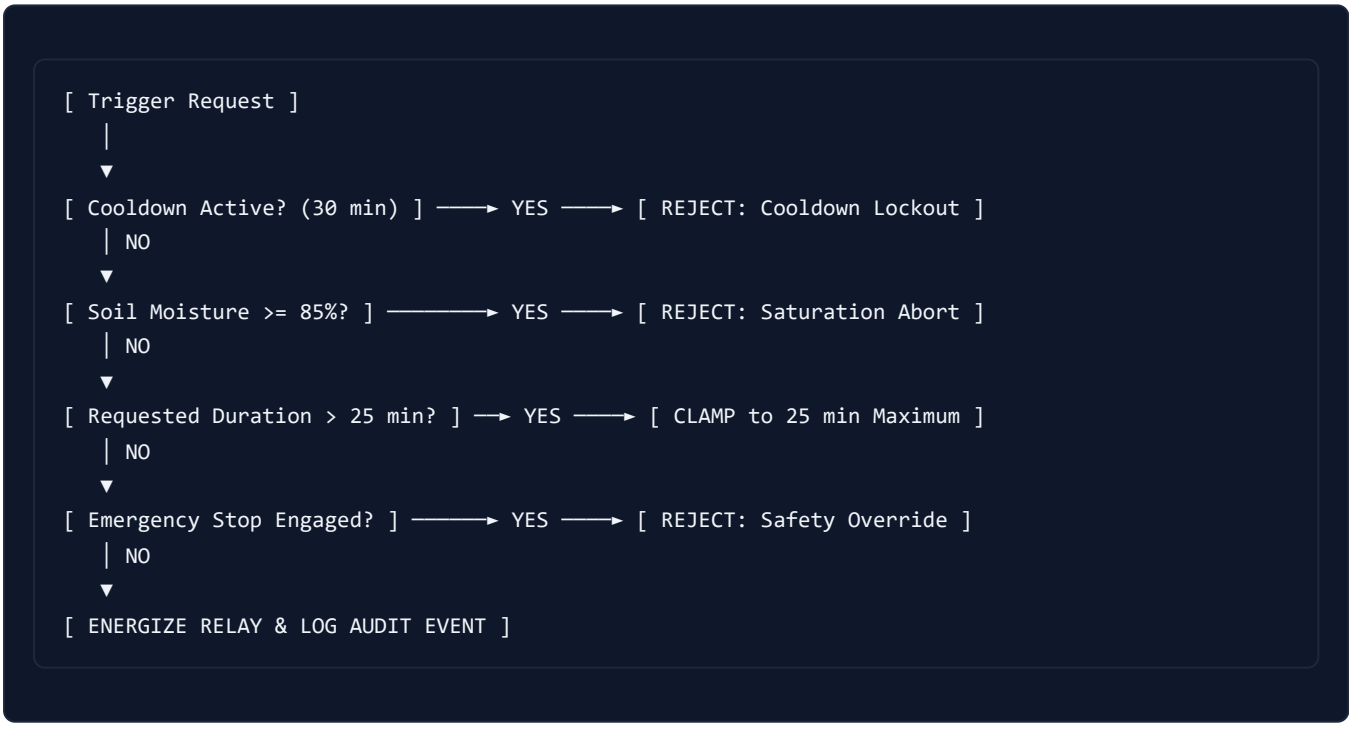
> **Equation:** `Volume_ML = Volume_SMD × (1 - ϕ_hydrogel)`

Where φ_hydrogel ∈ [0.25, 0.35] depending on soil organic carbon and texture.

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4. Pump Controller Safety Interlocks & Hardware Guardrails

The pump controller (`backend/services/pump_controller.py`) enforces strict safety interlocks before energizing the physical 12V submersible pump relay:



Safety Rules:

1. **Minimum Cooldown Lockout:** 30-minute mandatory rest period between successive pump pulses to permit water percolation and prevent over-saturation.
2. **Moisture Saturation Ceiling:** Hardware abort if sensor reading exceeds 85% moisture, preventing pump runaway.

3. **Hard Runtime Limit:** Maximum pulse duration clamped to 25 minutes (1,500 s).
4. **Emergency Stop:** Immediate physical or software override flipping relay low and writing a critical alert to `anomaly_events`.